Design & Implementation of a DBMS for Campus Facility Maintenance Request Management

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Problem Identification

Current Issues

Requests recorded manually; registers are fragmented and error-prone. Assignment and status tracking are inconsistent. Materials and cost records are incomplete.

Operational Impact

Delayed resolution · Poor accountability · Data redundancy · Limited performance insights for management decisions.

Scope & Objectives

Scope

Manage lifecycle of maintenance requests: logging → assignment → work execution → closure → feedback. Entities: buildings, rooms, assets, users, technicians, materials, costs.

Objectives

- Centralize maintenance data and histories
- Enable efficient registration, tracking, and assignment
- Record materials, costs, and technician work logs
- Enforce data integrity via constraints
- Produce management reports and analytics

Users & Roles

Student/Staff Requester

Raise requests, view status and provide post-resolution feedback.

Facility Office Staff

Validate requests, set priority, assign technicians and monitor progress.

Technician

View assignments, record work performed, materials used, and update status.

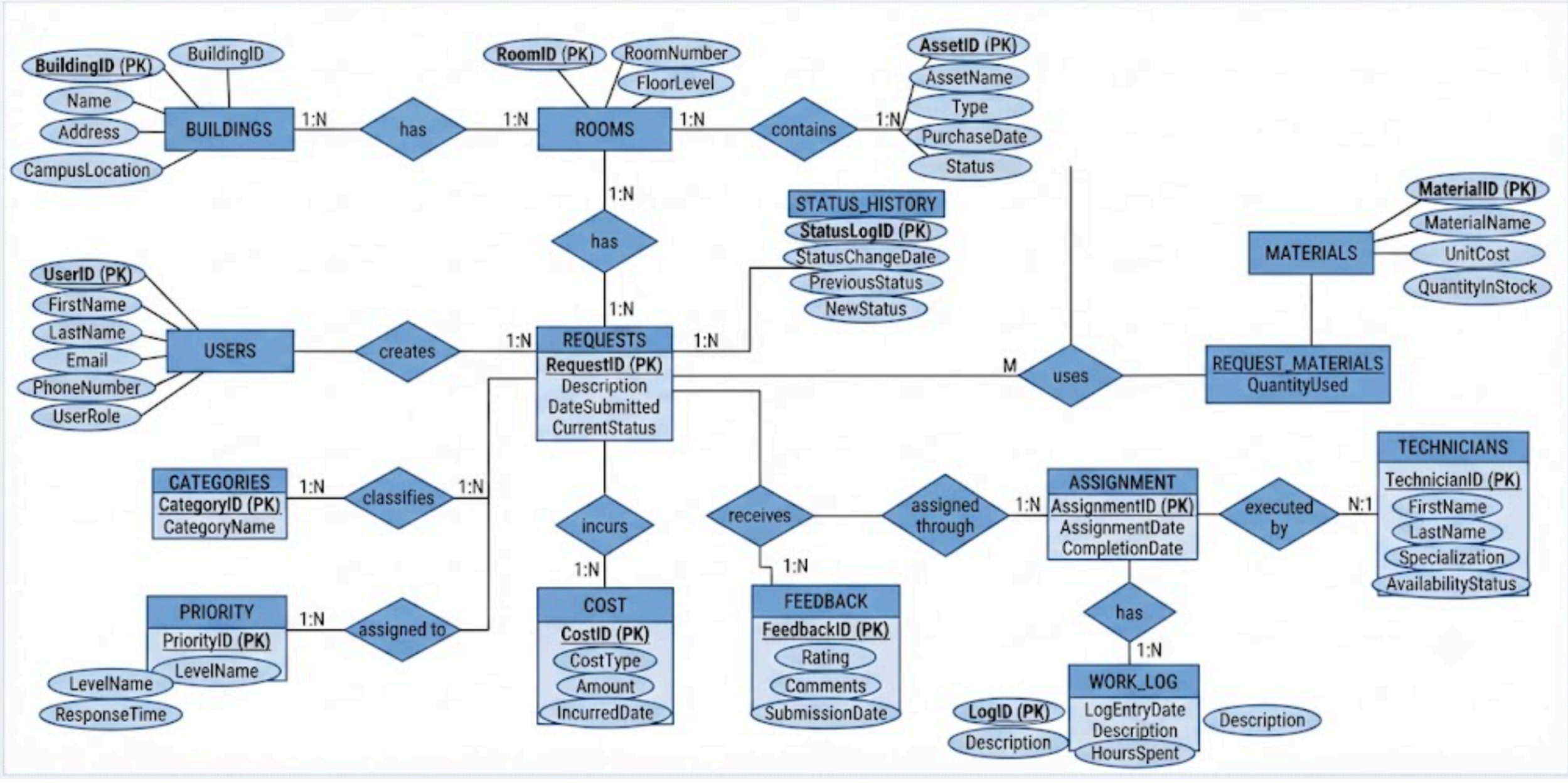
Storekeeper

Manage inventory, record material issues/usage tied to requests.

Admin / Management

Manage users/master data, monitor KPIs, and generate reports.

Entity Relationship Diagram – Campus Facility Maintenance System



Initial Relational Schema — Core Tables

Relation (Table)	Primary Key	Foreign Keys & Key Attributes
Buildings	`BuildingID` (PK)	BuildingName, CampusCode, TotalFloors
Rooms	`RoomID` (PK)	`BuildingID` (FK), RoomNumber, FloorNo
Assets	`AssetID` (PK)	`RoomID` (FK), `CategoryID` (FK), SerialNumber, Status
Users	`UserID` (PK)	FullName, Email, PasswordHash, Role (Student/Staff/Tech/Admin)
Requests	`RequestID` (PK)	`UserID` (FK), `AssetID` (FK), `PriorityID` (FK), Description

Initial Relational Schema — WorkFlow and Financials

Relation (Table)	Primary Key	Foreign Keys & Key Attributes
Technicians	`TechID` (PK)	`UserID` (FK), Specialization, AvailabilityFlag
Assignments	`AssignmentID` (PK)	`RequestID` (FK), `TechID` (FK), AssignedDate, Status
WorkLogs	`LogID` (PK)	`AssignmentID` (FK), HoursSpent, ActionNotes, LogDate
Materials & Costs	`CostID` (PK)	`WorkLogID` (FK), `MaterialID` (FK), QuantityUsed, TotalAmount
StatusHistories & Feedback	`HistoryID` / `FeedbackID`	`RequestID` (FK), PreviousStatus, NewStatus, Rating (1–5)

THANKYOU!